

DBS Lab 6 Tasks

1)

```

1  -- 1
2  SELECT * FROM EMPLOYEES
3  WHERE SALARY > (
4  SELECT AVG(SALARY) AS AVERAGE FROM EMPLOYEES
5  );
6

```

Query Result x

SQL | Fetched 50 rows in 0.007 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
6	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
7	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
8	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
9	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
10	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
11	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
12	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
13	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
14	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
15	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
16	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
17	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
18	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
19	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
20	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
21	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
22	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
23	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
24	153	Christopher	Olsen	COLSEN	011.44.1344.498718	30-MAR-06	SA_REP	8000	0.2	145	80
25	154	Nanette	Cambrault	NCAMBRAU	011.44.1344.987668	09-DEC-06	SA_REP	7500	0.2	145	80
26	155	Oliver	Tuvault	OTUVAULT	011.44.1344.486508	23-NOV-07	SA_REP	7000	0.15	145	80
27	156	Janette	King	JKING	011.44.1345.429268	30-JAN-04	SA_REP	10000	0.35	146	80
28	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
29	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80

2)

```

7  -- 2
8  SELECT * FROM EMPLOYEES
9  WHERE SALARY = (
10 SELECT MIN(e.SALARY) FROM EMPLOYEES e, DEPARTMENTS d
11 WHERE e.DEPARTMENT_ID = d.DEPARTMENT_ID
12 AND d.DEPARTMENT_NAME = 'IT'
13 );
14
15 -- 3

```

Query Result x

SQL | All Rows Fetched: 2 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
2	184	Nandita	Sarchand	NSARCHAN	650.509.1876	27-JAN-04	SH_CLERK	4200	(null)	121	50

3)

```

15  -- 3
16  SELECT * FROM EMPLOYEES
17  WHERE HIRE_DATE > ALL (
18  SELECT HIRE_DATE FROM EMPLOYEES
19  WHERE LAST_NAME = 'King'
20  );
21
22  -- 4

```

Query Result x

SQL | Fetched 50 rows in 0.01 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	192	Sarah	Bell	SBELL	650.501.1876	04-FEB-04	SH_CLERK	4000	(null)	123	50
2	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
3	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
4	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80
5	133	Jason	Mallin	JMALLIN	650.127.1934	14-JUN-04	ST_CLERK	3300	(null)	122	50
6	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
7	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80
8	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
9	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
10	142	Curtis	Davies	CDAVIES	650.121.2994	29-JAN-05	ST_CLERK	3100	(null)	124	50
11	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
12	131	James	Marlow	JAMRLOW	650.124.7234	16-FEB-05	ST_CLERK	2500	(null)	121	50
13	185	Alexis	Bull	ABULL	650.509.2876	20-FEB-05	SH_CLERK	4100	(null)	121	50
14	193	Britney	Everett	BEVERETT	650.501.2876	03-MAR-05	SH_CLERK	3900	(null)	123	50
15	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
16	159	Lindsey	Smith	LSMITH	011.44.1345.729268	10-MAR-05	SA_REP	8000	0.3	146	80
17	168	Lisa	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA_REP	11500	0.25	148	80
18	175	Alyssa	Hutton	AHUTTON	011.44.1644.429266	19-MAR-05	SA_REP	8800	0.25	149	80
19	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
20	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
21	188	Kelly	Chung	KCHUNG	650.505.1876	14-JUN-05	SH_CLERK	3800	(null)	122	50
22	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
23	125	Julia	Nayer	JNAYER	650.124.1214	16-JUL-05	ST_CLERK	3200	(null)	120	50
24	117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	2800	(null)	114	30
25	189	Jennifer	Dilly	JDILLY	650.505.2876	13-AUG-05	SH_CLERK	3600	(null)	122	50
26	202	Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK_REP	6000	(null)	201	20

4)

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22  -- 4
23  SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, DEPARTMENT_ID FROM EMPLOYEES
24  WHERE DEPARTMENT_ID = ANY (
25  SELECT DEPARTMENT_ID FROM DEPARTMENTS d, LOCATIONS l
26  WHERE d.LOCATION_ID = l.LOCATION_ID
27  AND l.CITY = 'Seattle'
28  );
29
30  -- 5

```

Query Result x

SQL | All Rows Fetched: 18 in 0.015 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
1	114	Den	Raphaely	30
2	115	Alexander	Khoo	30
3	116	Shelli	Baida	30
4	117	Sigal	Tobias	30
5	118	Guy	Himuro	30
6	119	Karen	Colmenares	30
7	108	Nancy	Greenberg	100
8	109	Daniel	Faviet	100
9	110	John	Chen	100
10	111	Ismael	Sciarra	100
11	112	Jose Manuel	Urman	100
12	113	Luis	Popp	100
13	100	Steven	King	90
14	101	Neena	Kochhar	90
15	102	Lex	De Haan	90
16	205	Shelley	Higgins	110
17	206	William	Gietz	110
18	200	Jennifer	Whalen	10

5)

```

30  -- 5
31  SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, DEPARTMENT_ID FROM EMPLOYEES
32  WHERE SALARY > ALL ( -- MAX SALARY IN DEPT 50 = 8200
33  SELECT SALARY FROM EMPLOYEES
34  WHERE DEPARTMENT_ID = 50
35  ); -- SELECTING > MAX(SALARY) IN DEPT 50 BETTER THAN THIS SUBQUERY
36
37  -- 6

```

Query Result x

SQL | All Rows Fetched: 31 in 0.005 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
1	206	William	Gietz	8300	110
2	177	Jack	Livingston	8400	80
3	176	Jonathon	Taylor	8600	80
4	175	Alyssa	Hutton	8800	80
5	103	Alexander	Hunold	9000	60
6	109	Daniel	Faviet	9000	100
7	152	Peter	Hall	9000	80
8	158	Allan	McEwen	9000	80
9	157	Patrick	Sully	9500	80
10	163	Danielle	Greene	9500	80
11	151	David	Bernstein	9500	80
12	170	Taylor	Fox	9600	80
13	204	Hermann	Baer	10000	70
14	169	Harrison	Bloom	10000	80
15	156	Janette	King	10000	80
16	150	Peter	Tucker	10000	80
17	162	Clara	Vishney	10500	80
18	149	Eleni	Zlotkey	10500	80
19	148	Gerald	Cambrault	11000	80
20	114	Den	Raphaely	11000	30
21	174	Ellen	Abel	11000	80
22	168	Lisa	Ozer	11500	80
23	147	Alberto	Errazuriz	12000	80
24	205	Shelley	Higgins	12008	110
25	108	Nancy	Greenberg	12008	100
26	201	Michael	Hartstein	13000	20

6)

```

37  -- 6
38  SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, DEPARTMENT_ID FROM EMPLOYEES
39  WHERE SALARY > ANY ( -- SALARY > MIN(SALARY) OF DEPT_ID 60 EMPS
40  SELECT SALARY FROM EMPLOYEES
41  WHERE DEPARTMENT_ID = 60
42  );
43
44  -- 7

```

Query Result x
SQL | Fetched 50 rows in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
1	100	Steven	King	24000	90
2	101	Neena	Kochhar	17000	90
3	102	Lex	De Haan	17000	90
4	145	John	Russell	14000	80
5	146	Karen	Partners	13500	80
6	201	Michael	Hartstein	13000	20
7	108	Nancy	Greenberg	12008	100
8	205	Shelley	Higgins	12008	110
9	147	Alberto	Errazuriz	12000	80
10	168	Lisa	Ozer	11500	80
11	114	Den	Raphaely	11000	30
12	148	Gerald	Cambrault	11000	80
13	174	Ellen	Abel	11000	80
14	149	Eleni	Zlotkey	10500	80
15	162	Clara	Vishney	10500	80
16	156	Janette	King	10000	80
17	150	Peter	Tucker	10000	80
18	204	Hermann	Baer	10000	70
19	169	Harrison	Bloom	10000	80
20	170	Taylor	Fox	9600	80
21	163	Danielle	Greene	9500	80
22	151	David	Bernstein	9500	80
23	157	Patrick	Sully	9500	80
24	152	Peter	Hall	9000	80
25	103	Alexander	Hunold	9000	60
26	158	Allan	McEwen	9000	80

7)

```

44 -- 7
45 SELECT * FROM EMPLOYEES
46 WHERE DEPARTMENT_ID IN (
47 SELECT DISTINCT DEPARTMENT_ID FROM EMPLOYEES
48 WHERE SALARY > 10000
49 );
50
51 -- 8

```

Query Result x

SQL | Fetched 50 rows in 0.009 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
5	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
6	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
7	111	Ismail	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
8	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
9	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
10	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
11	115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU_CLERK	3100	(null)	114	30
12	116	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU_CLERK	2900	(null)	114	30
13	117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	2800	(null)	114	30
14	118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2600	(null)	114	30
15	119	Karen	Colmenares	KCOLMENA	515.127.4566	10-AUG-07	PU_CLERK	2500	(null)	114	30
16	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
17	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
18	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
19	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
20	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
21	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
22	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
23	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
24	153	Christopher	Olsen	COLSEN	011.44.1344.498718	30-MAR-06	SA_REP	8000	0.2	145	80
25	154	Nanette	Cambrault	NCAMBRAU	011.44.1344.987668	09-DEC-06	SA_REP	7500	0.2	145	80
26	155	Oliver	Tuvault	OTUVAULT	011.44.1344.486508	23-NOV-07	SA_REP	7000	0.15	145	80

8)

```

51 -- 8
52 SELECT * FROM EMPLOYEES O
53 WHERE SALARY > (
54 SELECT AVG(SALARY) FROM EMPLOYEES I
55 WHERE O.DEPARTMENT_ID = I.DEPARTMENT_ID
56 );
57
58 -- 9

```

Query Result x

SQL | All Rows Fetched: 38 in 0.013 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
3	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
4	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
5	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
6	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
7	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
8	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
9	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
10	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
11	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
12	137	Renske	Ladwig	RLADWIG	650.121.1234	14-JUL-03	ST_CLERK	3600	(null)	123	50
13	141	Trenna	Rajs	TRAJS	650.121.8009	17-OCT-03	ST_CLERK	3500	(null)	124	50
14	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
15	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
16	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
17	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
18	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
19	150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	0.3	145	80
20	151	David	Bernstein	DBERNSTE	011.44.1344.345268	24-MAR-05	SA_REP	9500	0.25	145	80
21	152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	0.25	145	80
22	156	Janette	King	JKING	011.44.1345.429268	30-JAN-04	SA_REP	10000	0.35	146	80
23	157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	0.35	146	80
24	158	Allan	McEwen	AMCEWEN	011.44.1345.829268	01-AUG-04	SA_REP	9000	0.35	146	80
25	162	Clara	Vishney	CVISHNEY	011.44.1346.129268	11-NOV-05	SA_REP	10500	0.25	147	80
26	163	Danielle	Greene	DGREENE	011.44.1346.229268	19-MAR-07	SA_REP	9500	0.15	147	80

9)

```

58 -- 9
59 SELECT * FROM EMPLOYEES O
60 WHERE O.SALARY = (
61 SELECT MAX(I.SALARY) FROM EMPLOYEES I
62 WHERE I.DEPARTMENT_ID = O.DEPARTMENT_ID
63 ); -- 11 DISTINCT DEPT_IDS
64
65 -- 10

```

Query Result x

All Rows Fetched: 11 in 0.007 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
2	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
3	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
4	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
5	204	Hermann	Baer	HBAER	515.123.8888	07-JUN-02	PR_REP	10000	(null)	101	70
6	205	Shelley	Higgins	SHIGGINS	515.123.8080	07-JUN-02	AC_MGR	12008	(null)	101	110
7	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
8	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
9	203	Susan	Mavris	SMAVRIS	515.123.7777	07-JUN-02	HR_REP	6500	(null)	101	40
10	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
11	200	Jennifer	Whalen	JWHALEN	515.123.4444	17-SEP-03	AD_ASST	4400	(null)	101	10

10)

```

64
65 -- 10
66 SELECT * FROM DEPARTMENTS D
67 WHERE ( -- EMPLOYEE COUNT OF CURR DEPT
68 SELECT COUNT(E.DEPARTMENT_ID) FROM EMPLOYEES E
69 WHERE D.DEPARTMENT_ID = E.DEPARTMENT_ID
70 ) > ( -- AVG EMP COUNT PER DEPT
71 SELECT AVG(COUNT(*)) FROM EMPLOYEES
72 GROUP BY DEPARTMENT_ID
73 );
74

```

Query Result x

All Rows Fetched: 2 in 0.005 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	50	Shipping	121	1500
2	80	Sales	145	2500

11)

```

75  -- 11
76  SELECT O.JOB_ID, O.EMPLOYEE_ID, O.FIRST_NAME, O.LAST_NAME, O.SALARY
77  FROM EMPLOYEES O
78  WHERE O.SALARY > (
79  SELECT AVG(SALARY) FROM EMPLOYEES I
80  WHERE O.JOB_ID = I.JOB_ID
81  );
82
83  -- 12

```

Query Result x

SQL | All Rows Fetched: 43 in 0.008 seconds

	JOB_ID	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY
1	IT_PROG	103	Alexander	Hunold	9000
2	IT_PROG	104	Bruce	Ernst	6000
3	FI_ACCOUNT	109	Daniel	Faviet	9000
4	FI_ACCOUNT	110	John	Chen	8200
5	PU_CLERK	115	Alexander	Khoo	3100
6	PU_CLERK	116	Shelli	Baida	2900
7	PU_CLERK	117	Sigal	Tobias	2800
8	ST_MAN	120	Matthew	Weiss	8000
9	ST_MAN	121	Adam	Fripp	8200
10	ST_MAN	122	Payam	Kaufling	7900
11	ST_CLERK	125	Julia	Nayer	3200
12	ST_CLERK	129	Laura	Bissot	3300
13	ST_CLERK	130	Mozhe	Atkinson	2800
14	ST_CLERK	133	Jason	Mallin	3300
15	ST_CLERK	134	Michael	Rogers	2900
16	ST_CLERK	137	Renske	Ladwig	3600
17	ST_CLERK	138	Stephen	Stiles	3200
18	ST_CLERK	141	Trenna	Rajs	3500
19	ST_CLERK	142	Curtis	Davies	3100
20	SA_MAN	145	John	Russell	14000
21	SA_MAN	146	Karen	Partners	13500
22	SA_REP	150	Peter	Tucker	10000
23	SA_REP	151	David	Bernstein	9500
24	SA_REP	152	Peter	Hall	9000

12)

```

83  -- 12
84  SELECT * FROM EMPLOYEES
85  WHERE (JOB_ID, SALARY) IN ( -- EMPLOYEES WITH 'Taylor' in NAME
86  SELECT JOB_ID, SALARY FROM EMPLOYEES
87  WHERE 'Taylor' IN (FIRST_NAME, LAST_NAME)
88  );
89
90  -- 13
91  SELECT * FROM EMPLOYEES
92  WHERE (DEPARTMENT_ID, JOB_ID) IN (

```

Query Result x

SQL | All Rows Fetched: 3 in 0.007 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	180	Winston	Taylor	WTAYLOR	650.507.9876	24-JAN-06	SH_CLERK	3200	(null)	120	50
2	194	Samuel	McCain	SMCCAIN	650.501.3876	01-JUL-06	SH_CLERK	3200	(null)	123	50
3	176	Jonathon	Taylor	JTAYLOR	011.44.1644.429265	24-MAR-06	SA_REP	8600	0.2	149	80

13)


```

90 -- 13
91 SELECT * FROM EMPLOYEES
92 WHERE (DEPARTMENT_ID, JOB_ID) IN (
93 SELECT DEPARTMENT_ID, JOB_ID FROM EMPLOYEES
94 WHERE 'Whalen' IN (FIRST_NAME, LAST_NAME)
95 );
96
97 -- 14

```

Query Result x

SQL | All Rows Fetched: 1 in 0.005 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	200	Jennifer Whalen	JWHALEN	515.123.4444	17-SEP-03	AD_ASST	4400	(null)	101	10

14)

```

97 -- 14
98 SELECT * FROM DEPARTMENTS D
99 WHERE EXISTS (
100 SELECT 1 FROM EMPLOYEES E
101 WHERE E.DEPARTMENT_ID = D.DEPARTMENT_ID
102 );
103
104 -- 15

```

Query Result x

SQL | All Rows Fetched: 11 in 0.004 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	114	1700
4	40	Human Resources	203	2400
5	50	Shipping	121	1500
6	60	IT	103	1400
7	70	Public Relations	204	2700
8	80	Sales	145	2500
9	90	Executive	100	1700
10	100	Finance	108	1700
11	110	Accounting	205	1700

15)

```

104  -- 15
105  SELECT * FROM DEPARTMENTS D
106  WHERE NOT EXISTS (
107  SELECT 1 FROM EMPLOYEES E
108  WHERE E.DEPARTMENT_ID = D.DEPARTMENT_ID
109  );
110
111  -- 16

```

Query Result x

SQL | All Rows Fetched: 16 in 0.004 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	120	Treasury	(null)	1700
2	130	Corporate Tax	(null)	1700
3	140	Control And Credit	(null)	1700
4	150	Shareholder Services	(null)	1700
5	160	Benefits	(null)	1700
6	170	Manufacturing	(null)	1700
7	180	Construction	(null)	1700
8	190	Contracting	(null)	1700
9	200	Operations	(null)	1700
10	210	IT Support	(null)	1700
11	220	NOC	(null)	1700
12	230	IT Helpdesk	(null)	1700
13	240	Government Sales	(null)	1700
14	250	Retail Sales	(null)	1700
15	260	Recruiting	(null)	1700
16	270	Payroll	(null)	1700

16)

```
111  -- 16
112  SELECT * FROM EMPLOYEES
113  WHERE EMPLOYEE_ID IN (
114  SELECT EMPLOYEE_ID FROM JOB_HISTORY
115  );
116
117  -- 17
```

Query Result x

SQL | All Rows Fetched: 7 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
2	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
3	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
4	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
5	176	Jonathon	Taylor	JTAYLOR	011.44.1644.429265	24-MAR-06	SA_REP	8600	0.2	149	80
6	200	Jennifer	Whalen	JWHALEN	515.123.4444	17-SEP-03	AD_ASST	4400	(null)	101	10
7	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20

17)

```

117 -- 17
118 SELECT * FROM EMPLOYEES
119 WHERE EMPLOYEE_ID NOT IN (
120 SELECT EMPLOYEE_ID FROM JOB_HISTORY
121 );
122
123 -- 18

```

Query Result x

SQL | Fetched 50 rows in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
3	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
4	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
5	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
7	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
8	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
9	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
10	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
11	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
12	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100
13	115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU_CLERK	3100	(null)	114	30
14	116	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU_CLERK	2900	(null)	114	30
15	117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	2800	(null)	114	30
16	118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2600	(null)	114	30
17	119	Karen	Colmenares	KCOLMENA	515.127.4566	10-AUG-07	PU_CLERK	2500	(null)	114	30
18	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
19	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
20	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
21	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
22	125	Julia	Nayer	JNAYER	650.124.1214	16-JUL-05	ST_CLERK	3200	(null)	120	50
23	126	Irene	Mikkilineni	IMIKKILI	650.124.1224	28-SEP-06	ST_CLERK	2700	(null)	120	50
24	127	James	Landry	JLANDRY	650.124.1334	14-JAN-07	ST_CLERK	2400	(null)	120	50

18)

```

123 -- 18
124 SELECT * FROM (
125 SELECT EMPLOYEE_ID, FIRST_NAME, SALARY, DEPARTMENT_ID
126 FROM EMPLOYEES
127 ORDER BY SALARY DESC)
128 WHERE ROWNUM <= 3;
129
130 -- 19
131 SELECT DEPARTMENT_ID, AVG(SALARY) AS AVG_SALARY

```

Query Result x

SQL | All Rows Fetched: 3 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	SALARY	DEPARTMENT_ID
1	100	Steven	24000	90
2	101	Neena	17000	90
3	102	Lex	17000	90

19)

[illegible]

20)

```
138 -- 20
139 CREATE TABLE HIGH_PAID_EMPLOYEES AS
140 SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, DEPARTMENT_ID
141 FROM EMPLOYEES
142 WHERE 1 = 0 -- COPY ONLY STRUCTURE, NO ROWS
143 ;
144
145 INSERT INTO HIGH_PAID_EMPLOYEES
146     (EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, DEPARTMENT_ID)
147 SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, DEPARTMENT_ID
148 FROM EMPLOYEES
149 WHERE SALARY > (
150 SELECT AVG(SALARY) FROM EMPLOYEES
151 );
152
153 SELECT * FROM HIGH_PAID_EMPLOYEES;
154
```

Query Result x Script Output x
Task completed in 0.044 seconds

Table HIGH_PAID_EMPLOYEES created.

51 rows inserted.

152

153

154

155

-- 21

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.009 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
1	100	Steven	King	24000	90
2	101	Neena	Kochhar	17000	90
3	102	Lex	De Haan	17000	90
4	103	Alexander	Hunold	9000	60
5	108	Nancy	Greenberg	12008	100
6	109	Daniel	Faviet	9000	100
7	110	John	Chen	8200	100
8	111	Ismael	Sciarra	7700	100
9	112	Jose Manuel	Urman	7800	100
10	113	Luis	Popp	6900	100
11	114	Den	Raphaely	11000	30
12	120	Matthew	Weiss	8000	50
13	121	Adam	Fripp	8200	50
14	122	Payam	Kaufling	7900	50
15	123	Shanta	Vollman	6500	50
16	145	John	Russell	14000	80

21)

```
155 -- 21
156 UPDATE EMPLOYEES O
157 SET SALARY = SALARY + (SALARY * 0.1)
158 WHERE SALARY < (
159 SELECT AVG(SALARY) FROM EMPLOYEES I
160 WHERE I.DEPARTMENT_ID = O.DEPARTMENT_ID
161 );
162
163 -- 22
164 DELETE FROM EMPLOYEES
```

Script Output x Query Result x
Task completed in 0.063 seconds

Table HIGH_PAID_EMPLOYEES created.

51 rows inserted.

65 rows updated.

22)

```

163  -- 22
164  DELETE FROM EMPLOYEES
165  WHERE DEPARTMENT_ID = ANY (
166      SELECT DEPARTMENT_ID FROM DEPARTMENTS D, LOCATIONS L
167      WHERE D.LOCATION_ID = L.LOCATION_ID
168      AND L.CITY = 'Toronto'
169  );
170

```

Script Output x Query Result x

Task completed in 0.047 seconds

65 rows updated.

Error starting at line : 164 in command -

```

DELETE FROM EMPLOYEES
WHERE DEPARTMENT_ID = ANY (
    SELECT DEPARTMENT_ID FROM DEPARTMENTS D, LOCATIONS L
    WHERE D.LOCATION_ID = L.LOCATION_ID
    AND L.CITY = 'Toronto'
)
Error report -
ORA-02292: integrity constraint (HR.DEPT_MGR_FK) violated - child record found

```

23)

```

171  -- 23
172  SELECT * FROM EMPLOYEES
173  WHERE SALARY > (
174      SELECT AVG(AVG_DEPT_SALARY)
175      FROM (
176          SELECT AVG(SALARY) AS AVG_DEPT_SALARY
177          FROM EMPLOYEES
178          GROUP BY DEPARTMENT_ID
179          HAVING AVG(SALARY) > (
180              SELECT AVG(SALARY) FROM EMPLOYEES
181          )
182      )
183  );
184
185  -- 24
186  SELECT * FROM EMPLOYEES

```

Script Output x Query Result x

All Rows Fetched: 13 in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	18700	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	18700	(null)	100	90
4	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
5	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
6	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	0.4	100	80
7	146	Karen	Partners	KPARTNER	011.44.1344.467268	05-JAN-05	SA_MAN	13500	0.3	100	80
8	147	Alberto	Errazuriz	AERRAZUR	011.44.1344.429278	10-MAR-05	SA_MAN	12000	0.3	100	80
9	148	Gerald	Cambrault	GCAMBRAU	011.44.1344.619268	15-OCT-07	SA_MAN	11000	0.3	100	80
10	168	Lisa	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA_REP	11500	0.25	148	80
11	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80
12	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
13	205	Shelley	Higgins	SHIGGINS	515.123.8080	07-JUN-02	AC_MGR	12008	(null)	101	110

24)


```

185 -- 24
186 SELECT * FROM EMPLOYEES
187 WHERE DEPARTMENT_ID IN (
188     SELECT DEPARTMENT_ID FROM DEPARTMENTS
189     WHERE LOCATION_ID IN (
190         SELECT LOCATION_ID FROM LOCATIONS
191         WHERE CITY IN (
192             SELECT L2.CITY
193             FROM EMPLOYEES E2, DEPARTMENTS D2, LOCATIONS L2
194             WHERE E2.DEPARTMENT_ID = D2.DEPARTMENT_ID
195             AND D2.LOCATION_ID = L2.LOCATION_ID
196             GROUP BY L2.CITY
197             HAVING AVG(E2.SALARY) > (
198                 SELECT AVG(SALARY) FROM EMPLOYEES
199             )
200         )
201     )
202 );
203

```

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
2	115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU_CLERK	3410	(null)	114	30
3	116	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU_CLERK	3190	(null)	114	30
4	117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	3080	(null)	114	30
5	118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2860	(null)	114	30
6	119	Karen	Colmenares	KCOLMENA	515.127.4566	10-AUG-07	PU_CLERK	2750	(null)	114	30
7	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
8	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
9	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	9020	(null)	108	100
10	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	8470	(null)	108	100
11	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	8580	(null)	108	100
12	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	7590	(null)	108	100
13	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
14	202	Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK_REP	6600	(null)	201	20
15	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90

25)

```

204 -- 25
205 SELECT * FROM EMPLOYEES
206 WHERE SALARY >= ( -- '=' USED BCUZ ONE CANT BE GREATER THAN ONESELF
207     SELECT MAX(SALARY)
208     FROM EMPLOYEES
209     WHERE DEPARTMENT_ID IN (
210         SELECT DEPARTMENT_ID
211         FROM EMPLOYEES
212         GROUP BY DEPARTMENT_ID
213         HAVING COUNT(EMPLOYEE_ID) < 1
214         SELECT AVG(COUNT(EMPLOYEE_ID))
215         FROM EMPLOYEES
216         GROUP BY DEPARTMENT_ID
217     )
218 );
219
220

```

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90